

Sparse State Estimation under Sensor Corruption

Gokul Rajaraman (22B4517) and Rohan Badgujar (22B2196)

Linear System Model

Simulation model

$$x_{k+1} = Ax_k + Bu_k + w_k$$

$$y_k = Cx_k + e_k + v_k$$

Noise model

$$w_k \sim \mathcal{N}(0, \sigma_w^2 I), \quad v_k \sim \mathcal{N}(0, \sigma_v^2 I)$$

Experiment setup

- ▶ State dimension: $n = 12$
- ▶ Output dimension: $p = 6 = n/2$
- ▶ Time horizon: $T = 400$
- ▶ Number of corrupted sensors: $s = 2$

System Structure Used in the Code

Dynamics matrix

- ▶ A is a stable tridiagonal coupling matrix
- ▶ diagonal entries dominate off-diagonal coupling
- ▶ this keeps the open-loop trajectory bounded

Input matrix

$$B = [1, 0, \dots, 0]^T$$

Observation matrix

$$C = [I_6 \quad 0]$$

- ▶ only the first $n/2$ states are measured directly
- ▶ the remaining states must be inferred through coupling in A

Control law used in simulation

$$u_k = -Kx_k, \quad K = 0.2I$$

Sparse Sensor Corruption Model

Corrupted measurement model

$$y_k = Cx_k + e_k + v_k$$

Corruption structure in the experiment

- ▶ only $s = 2$ sensors are corrupted
- ▶ the corrupted sensor indices are fixed over time
- ▶ corruption is constant over time for each selected sensor

Equivalent representation

$$e_k = e$$

for all k , with e sparse in the sensor dimension.

Interpretation

- ▶ this models a persistent sensor bias
- ▶ it is closer to a faulty sensor than a transient spike

Batch Estimation Problem

Optimization used for recovery

$$\min_{x_{0:T}, e_{0:T}} \sum_{k=0}^T \|y_k - Cx_k - e_k\|_2^2 + \lambda \|e_k\|_1$$

Estimator dynamics in the code

$$x_{k+1} = Ax_k$$

Important modeling note

- ▶ the batch estimator uses the nominal dynamics only
- ▶ the known input term Bu_k is included in simulation, but not enforced in the optimization
- ▶ this keeps the recovery problem convex and easy to solve with CVXPY

Regularization Parameter Sweep

New experiment

- ▶ sweep the regularization parameter λ on a log scale
- ▶ measure both corruption recovery error and state estimation error
- ▶ use the same system, observation structure, and sparse bias model

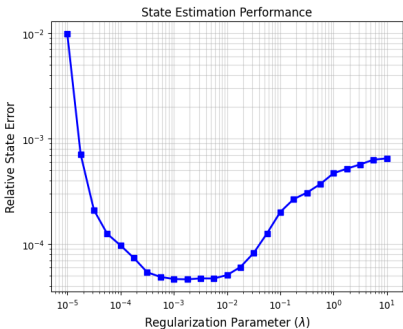
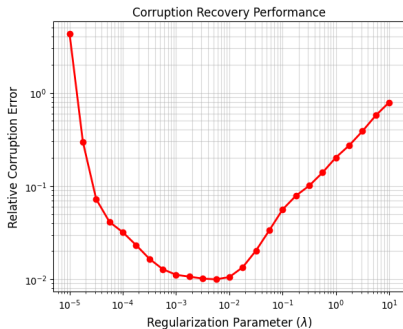
Evaluation metrics

$$\frac{\|e_{\text{true}} - e_{\text{estimated}}\|_2}{\|e_{\text{true}}\|_2}, \quad \frac{\|x_{\text{true}} - x_{\text{estimated}}\|_2}{\|x_{\text{true}}\|_2}$$

Core idea

- ▶ small λ gives weak sparsity pressure
- ▶ large λ over-penalizes the corruption term
- ▶ the best performance appears at an intermediate λ

Corruption Recovery & State Estimation vs Regularization

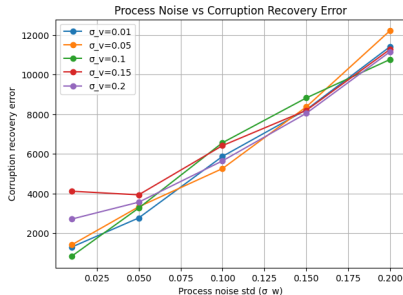


Corruption Recovery & State Estimation vs Regularization

Interpretation

- ▶ both corruption and state errors show a U-shaped trend versus λ
- ▶ very small λ under-regularizes, while very large λ suppresses the corruption term
- ▶ an intermediate λ balances sparsity and data fit, giving the best performance
- ▶ state error remains smaller than corruption error, indicating the dynamics help stabilize estimation

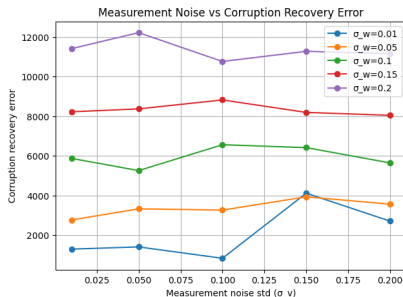
Process Noise vs Corruption Recovery Error



Observed trend

- ▶ corruption recovery error increases as σ_w grows
- ▶ stronger process noise makes the state trajectory harder to predict
- ▶ the mismatch gets partly absorbed into the estimated corruption

Measurement Noise vs Corruption Recovery Error



Observed trend

- ▶ higher σ_v generally leads to worse corruption recovery
- ▶ the curve is not perfectly monotone because each point is one random trial
- ▶ at larger noise levels, distinguishing v_k from sparse e_k becomes harder

Interpretation and Takeaway

Why recovery changes across the plots

- ▶ λ controls the tradeoff between data fit and sparsity
- ▶ process noise perturbs the state evolution
- ▶ measurement noise masks the sparse corruption signature
- ▶ only half the states are observed directly

Takeaway

Moderate $\lambda \Rightarrow$ best corruption recovery

Lower $\sigma_w, \sigma_v \Rightarrow$ better sparse recovery

Higher $\sigma_w, \sigma_v \Rightarrow$ more ambiguity between noise and attack

Nonlinear Benchmark Model

Coupled damped pendulum network

For a chain of N pendulums, define the state as

$$x_k = \begin{bmatrix} \theta_k \\ \omega_k \end{bmatrix} \in \mathbb{R}^{2N}$$

where

- ▶ $\theta_k \in \mathbb{R}^N$ are the joint angles
- ▶ $\omega_k \in \mathbb{R}^N$ are the angular velocities

Measurement model

$$y_k = Cx_k + e_k + v_k$$

with sparse sensor corruption e_k and measurement noise v_k .

Nonlinear Governing Equations

Discrete-time dynamics used in the code

$$\theta_{k+1} = \theta_k + dt \omega_k$$

$$\omega_{k+1} = \omega_k + dt \left(-a \sin(\theta_k) - d \omega_k + c L \theta_k \right)$$

Meaning of the terms

- ▶ $-a \sin(\theta_k)$ is the nonlinear restoring torque
- ▶ $-d \omega_k$ is viscous damping
- ▶ $c L \theta_k$ is nearest-neighbor coupling through the graph Laplacian

Laplacian Coupling

1-D chain Laplacian

For $N = 5$, the matrix has the form

$$L = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Interior node coupling

$$(L\theta)_i = \theta_{i-1} - 2\theta_i + \theta_{i+1}$$

Boundary nodes

$$(L\theta)_1 = \theta_2 - \theta_1, \quad (L\theta)_N = \theta_{N-1} - \theta_N$$

Physical Interpretation

- ▶ each pendulum interacts only with its neighbors
- ▶ coupling depends on angle differences, not absolute values
- ▶ the Laplacian acts like a discrete second spatial derivative
- ▶ motion in one pendulum propagates through the chain

Why this is a good benchmark

- ▶ nonlinear
- ▶ sparse and local coupling
- ▶ partially observed
- ▶ corruption in one sensor affects estimation across the network

Nonlinear Estimation Problem

Batch formulation

$$\min_{x_{0:T}, e_{0:T}} \sum_{k=0}^T \|y_k - Cx_k - e_k\|_2^2 + \lambda \|e_k\|_1$$

Subject to nonlinear dynamics

$$x_{k+1} = f(x_k)$$

where $f(\cdot)$ is the pendulum network dynamics.

Solver strategy

- ▶ OSQP: solve the linearized subproblem iteratively
- ▶ IPOPT: solve the nonlinear constrained problem directly

Linearization Used for OSQP

Local approximation around the current iterate $\bar{x}_k$

$$f(x_k) \approx f(\bar{x}_k) + J_f(\bar{x}_k)(x_k - \bar{x}_k)$$

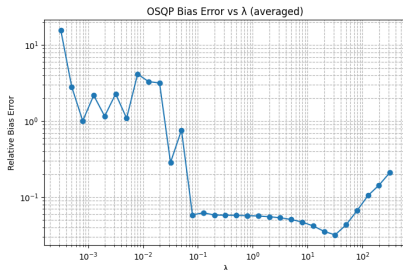
Resulting convex subproblem

$$\min_{x,e} \sum_k \|y_k - Cx_k - e_k\|_2^2 + \lambda \|e_k\|_1$$

$$\text{s.t. } x_{k+1} = f(\bar{x}_k) + J_f(\bar{x}_k)(x_k - \bar{x}_k)$$

- ▶ repeated until convergence
- ▶ keeps the sparse recovery step computationally tractable

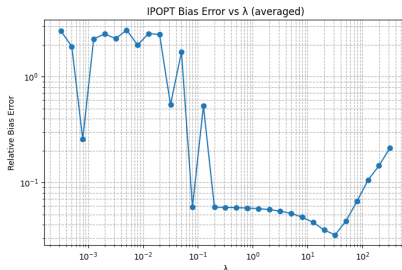
OSQP Bias Error vs Regularization Parameter



Interpretation

- ▶ the curve is U-shaped on the log scale
- ▶ very small λ under-regularizes, so bias is not isolated cleanly
- ▶ very large λ suppresses the bias term too much
- ▶ the best recovery occurs at an intermediate λ

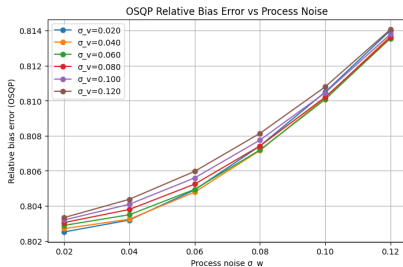
IPOPT Bias Error vs Regularization Parameter



Interpretation

- ▶ IPOPT shows the same overall U-shaped trend
- ▶ moderate λ gives the best tradeoff between fit and sparsity
- ▶ the nonlinear solver is more stable near the optimum
- ▶ both solvers agree on the need for intermediate regularization

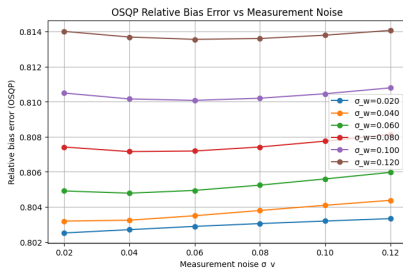
Relative Bias Error vs Process Noise



Interpretation

- ▶ relative bias error increases slowly as process noise grows
- ▶ the curves are close together, so measurement noise has a weaker effect here
- ▶ larger process noise makes the nonlinear state prediction less reliable
- ▶ the estimator then absorbs some mismatch into the bias term

Relative Bias Error vs Measurement Noise



Interpretation

- ▶ relative bias error increases gradually as measurement noise σ_v grows
- ▶ higher measurement noise makes corrupted and healthy sensors harder to distinguish
- ▶ curves remain smooth and nearly parallel, indicating stable estimator behavior
- ▶ the overall sensitivity to measurement noise is moderate in this regime

Nonlinear Results Summary

- ▶ the pendulum network is a coupled nonlinear benchmark with sparse local interactions
- ▶ Laplacian coupling spreads motion through the chain through angle differences
- ▶ bias recovery is best at moderate λ
- ▶ both OSQP and IPOPT show the same sparsity tradeoff
- ▶ increasing process noise worsens bias recovery, but only gradually in this regime

Takeaway

Nonlinearity + coupling + sparse corruption



Harder but more realistic estimation